



NNP Pocket Guide 2

FEN, CV, and Resp

Version 1, October 2026

Companion to the procedures booklet; same ring.

Tags

Course

N5537 lectures and handouts (Dr. Short), and your study guides built from them

Card

Your instructor's badge card

Dir.

Program director's NNP Brain

AAP

AAP 2022 hyperbilirubinemia guideline



Sources disagree, or an error was found

Sections



FEN: GIR, TPN, calories, bilirubin



CV: pressors, profiles, BP norms



Resp: blood gas, vent



Ref: sources

Starting points, not orders. Follow unit protocol and your attending.



GIR (mg/kg/min)

Shortest	$\%D \times \text{mL/kg/day} \div 144$ Course
Hourly	$\%D \times \text{mL/hr} \div (6 \times \text{kg})$ Course Dir. $= \%D \times \text{mL/hr} \times 0.167 \div \text{kg}$ Card
Long way	$\text{mL/day} \times \%D \text{ (decimal)} = \text{g}; \times 1000 = \text{mg}; \div \text{kg} \div 1440$ Course

%D for a target GIR = $\text{GIR} \times 144 \div \text{mL/kg/day}$

Example: D10 at 80 mL/kg/day = GIR 5.6.

Targets

Start 4–6, then titrate to glucose; preterm need at least 4.

[Course](#)

Target 4–8. [Card](#)

Intro lecture: 2–4 most newborns, 8–10 SGA/LGA, >15 IDM.

[Course](#)

Max dextrose [Course](#)

12.5% peripheral. 25% central with the tip in the SVC.



TPN: start and advance

When: <32 wk, VLBW/ELBW, GI anomaly or surgery, NPO >3–5 days. [Course](#)

Day 1: D10 at 60–80 mL/kg/d, AA 2–3 g/kg/d, lipid 1 g/kg/d.

[Course](#)

Amino acids	Within 24 h. AA slide: start 3, +0.5/d to 3–4, max 4. Course Card: start 2–3, goal 3–4. Card
Lipid	1–2 → 3 g/kg/d. 0.5–1 prevents EFAD. TG >200: stop or cut 0.5–1. Course
Sodium	None first 24 h, then 2–4 Course Card: 1–2 from day 1, goal 3–4 Card
Potassium	None first 24–48 h, then 1–3 once urine output is established Course Card: 1–2 from day 1, goal 2–4 Card
Ca gluconate	200–400 mg/kg/d Course Card

The lecture writes Na and K as mEq/d; they're usually written per kg per day.

Peripheral TPN under 900 mOsm/L. [Course](#)



TPN calories

Glucose	3.4 kcal/g Course Dir. (card: carbohydrate 4)
Protein	4 kcal/g
Fat	9 kcal/g
20% lipid	1 g = 5 mL = 10 kcal (2 kcal/mL) Course

$$\text{Total} = \text{GIR} \times 5 + \text{AA g/kg} \times 4 + \text{lipid g/kg} \times 10 \quad \text{Course}$$

Exact dextrose kcal/kg/day = %D $\times$ mL/kg/day $\times$ 0.034 (GIR $\times$ 4.9).

$$\text{Lipid g/kg/day} = \text{mL/hr} \times 24 \times 0.2 \div \text{kg} \quad \text{Course}$$

Example: 1.8 kg, D12 at 10.2 mL/hr, AA 3.5, lipid 1.1 mL/hr $\approx$ 99 kcal/kg/day.

Goals [Course](#)

Preterm 110–130, term 100–120 kcal/kg/day. Gain 15–20 g/kg/day.

Feeds: kcal/mL = kcal/oz $\div$ 30

20 $\rightarrow$ 0.67	22 $\rightarrow$ 0.73	24 $\rightarrow$ 0.8
26 $\rightarrow$ 0.87	28 $\rightarrow$ 0.93	30 $\rightarrow$ 1.0



Bilirubin

35 weeks and up AAP

Thresholds are hour-specific curves by GA and risk: plot them in BiliTool or PediTools.

Neurotoxicity risk factors: GA <38 wk, albumin <3.0, hemolysis (isoimmune, G6PD), sepsis, instability in the past 24 h.

Escalation: TSB within 2 of the exchange line = emergency. NICU, intensive phototherapy, PO + IV fluids, STAT labs, TSB every 2 h.

Stop phototherapy at ≥ 2 below the start threshold; rebound in ~24 h (6–12 h if started before 48 h).

Under 35 weeks (Maisels 2012)

PMA	Photo	Exchange
<28 wk	5–6	11–14
28–29	6–8	12–14
30–31	8–10	13–16
32–33	10–12	15–18
34–35	12–14	17–19

TSB mg/dL. Born <35 wk: stay on the top row after 35 wk PMA.



Pick the pressor

BP = CO × SVR: pipes, pump, or both?

Low diastolic	Pipes (low SVR) → vasopressor ($\alpha 1$ or V1)
Low systolic	Pump (low contractility) → inotrope ($\beta 1$ or PDE3)
Both	Cold, mottled, acidotic → both actions

	Warm (low SVR)	Cold (high SVR)
Skin	Warm, pink	Cool, mottled
Refill	Flash <2 s	>3 s
Pulses	Bounding	Thready
Pulse pressure	Wide	Narrow
Causes	Sepsis, NEC, big PDA	Cold sepsis, transitional, post-ligation

Hypotensive, warm, and bounding is **not** reassuring: it's vasodilated and needs tone. [Course](#)



Pressor doses

Drug	Dose	Use
Dopamine	≤ 2 / $\geq 5-10$ / ≥ 10 mcg/kg/min	DA $\rightarrow$ $\beta_1 \rightarrow \alpha_1$ with dose. ≥ 10 : R $\rightarrow$ L ductal shunt risk. Not with PPHN.
Dobutamine	$\geq 3-15$ mcg/kg/min	β_1 inotrope, mild β_2 dilation. Avoid in outflow obstruction (IDM).
Epinephrine	0.02–0.1, then ≥ 0.1 (caution >0.2)	Low dose β , high dose α_1 . Lactate, glucose, tachycardia.
Norepinephrine	0.02–0.4 mcg/kg/min	$\alpha_1 > \beta_1$. Severe vasoplegia.
Vasopressin	0.0002–0.005 units/kg/min	V1a. Catecholamine-resistant; spares PVR. Low Na, low platelets.
Milrinone	0.25–0.75 mcg/kg/min, no load	PDE3 inodilator. Low output + PH, post-ligation. Onset 1–2 h; drops BP.
Hydrocortisone	1 mg/kg IV q8h	Adrenal insufficiency, or refractory on ≥ 2 agents.

Practice-question guide uses dopamine tiers 1–5 / 5–10 / >10 . [Course](#)



Decision tree and algorithm

1. **Bed loose** (warm, low diastolic) → norepinephrine (or dopamine)
2. **Pump weak** (cool, poor contractility) → dobutamine
3. **Pump weak + high afterload** (PPHN, post-op) → milrinone, add a pressor if BP sags
4. **Both failing** → epinephrine
5. **Still low on catecholamines** → vasopressin; think hydrocortisone

Around it Course

Perfusion adequate (NIRS, urine output, lactate, echo)? Then permissive hypotension. Reassess after every change; BP up with worse perfusion means over-constriction. Wean catecholamines first, keep milrinone while PH persists, wean vasopressin last.

► The lecture defaults to dopamine first for vasoplegia; much current practice starts phenotype-specific (e.g., norepinephrine or epinephrine in septic shock). Practice question 5 keys dopamine for high-SVR cold shock against the lecture's own afterload logic. Know the keyed answers; follow your unit.



Hemodynamic profiles

Read diastolic BP, pulse pressure, perfusion index, SVR, cardiac output, and StO₂ together. [Course](#)

Profile	Pattern
Vasodilatory	↓ DBP, ↓ SVR; ↑ PP, PI, CO (warm shock)
Vasoconstrictor	↑ DBP, ↑ SVR; ↓ PP, PI, CO (cold shock)
Cardiogenic	↓ CO, low PP, PI, DBP
L-R shunt	↑ CO, ↑ PP; ↓ DBP, ↓ SVR (PDA run-off)
Volume depletion	↓ CO, ↓ PP; ↑ SVR

Same low diastolic, opposite fixes: vasodilatory needs a vasoconstrictor; volume-depleted needs volume.

Epinephrine 0.2 mcg/kg/min for an hour $\approx$ one resuscitation bolus (10 mcg/kg).



BP norms (after 2 weeks)

By PMA; 50th / 95th / 99th percentile, mmHg (Dionne 2012, corrected table).

PMA	MAP	Systolic	Diastolic
44	63/80/85	88/105/110	50/68/73
42	62/76/81	85/98/102	50/65/70
40	60/75/80	80/95/100	50/65/70
38	59/74/79	77/92/97	50/65/70
36	57/72/77	72/87/92	50/65/70
34	50/65/70	70/85/90	40/55/60
32	49/64/69	68/83/88	40/55/60
30	48/63/68	65/80/85	40/55/60
28	45/58/63	60/75/80	38/50/54
26	38/57/63	55/72/77	30/50/56

High BP: $\geq$ 95th on 3 occasions; $>$ 99th, investigate. "**MAP** $\geq$ **GA**": weak evidence; judge with perfusion. **Cuff:** compare MAP (the measured value), right arm.



Read a blood gas

1. **pH:** <7.35 acidemic, >7.45 alkalemic
2. **PaCO_2 :** respiratory?
3. **HCO_3^- / **BE:** metabolic?**
4. **Compensation, then the bedside picture**

CO_2 opposite pH $\rightarrow$ respiratory. HCO_3^- with pH $\rightarrow$ metabolic.

pH	7.35–7.45 (>7.30 may be OK in context)
PaCO_2	35–45
HCO_3^-	22–26
BE	-2 to +2; worse than -4 to -6, watch

Uncompensated	pH off, other value normal
Partial	pH off, both abnormal
Compensated	pH normal, both abnormal

Lungs fix CO_2 quickly; kidneys fix HCO_3^- over days. High CO_2 **and** low HCO_3^- = mixed acidosis, not compensation. [Course](#)



What moves what

CO₂ ↑	pH ↓ (respiratory acidosis)
CO₂ ↓	pH ↑ (respiratory alkalosis)
HCO₃⁻ ↓ (lactate, TPN acid, RTA, diarrhea)	pH ↓, BE more negative
HCO₃⁻ ↑ (vomiting, diuretics, bicarb, acetate)	pH ↑
Bicarb given	HCO ₃ ⁻ + H ⁺ → carbonic acid → CO₂ + water : CO ₂ rises; pH can fall if not ventilated
CO₂ high for days	Kidneys keep HCO ₃ ⁻ → pH toward normal
Metabolic acidosis	Lungs blow off CO ₂ quickly
Rate or tidal volume ↑	CO ₂ ↓, pH ↑
MAP or FiO₂ ↑	Oxygenation ↑

Bicarb: only with adequate ventilation; not in the same line as catecholamines; can lower K.

► The Pat Thomas guide says kidneys "retain H⁺" in respiratory acidosis; they excrete H⁺ and keep HCO₃⁻.



Vent modes

AC	Every breath supported. Poor effort, apnea, extreme prematurity.
SIMV	Synced breaths plus spontaneous. Weaning, low support, sedation backup.
Volume guarantee	Hits a set volume; unstable compliance; guards against volutrauma.
Pressure control	Caps pressure; volume varies. RDS, severe BPD.
HFOV	Open lung, tiny volumes. PIE, air leak, PPHN.
HFJV	Fast small bursts; conventional vent gives PEEP. PIE, refractory CO ₂ .

By disease [Course](#)

RDS: CPAP → PC or volume-targeted; HFOV if severe. **TTN:** HFNC or CPAP. **BPD:** long expiratory time, permissive CO₂, often VG. **PIE/air leak:** HFJV or HFOV. **Pulmonary hemorrhage:** higher MAP. **PPHN:** conventional or HFOV.



What to change: conventional and HFOV

Conventional

CO₂ high	Rate +5–10; PIP +2 for chest rise; lower PEEP if over-expanded
O₂ low	FiO ₂ ↑; PEEP ↑; PIP +1–2; Ti 0.4–0.6 s
Weaning	FiO ₂ to 80% before changing PIP, I:E, PEEP; steps PIP 2, FiO ₂ 5%, rate 5

$$\text{MAP} = (\text{PIP} - \text{PEEP}) \times \text{Ti} \div (\text{Ti} + \text{Te}) + \text{PEEP}$$

HFOV

CO₂ high	Amplitude ↑; lower Hz if needed
CO₂ low	Amplitude ↓; raise Hz
O₂ low	MAP ↑; FiO ₂ ↑
O₂ high	FiO ₂ ↓, then MAP ↓

Lower Hz = bigger volumes, more CO₂ removal; Hz changes are usually a senior call.



Jet, weaning, and OI

HFJV

Ventilation: PIP above PEEP (ΔP). Oxygenation: PEEP (+1–2 when $FiO_2 \geq 0.6$). Don't wean PEEP until $FiO_2 < 0.35$ – 0.40 .

HFOV weaning Dir.

- $PaCO_2 < 50$ and $pH > 7.25$: wean amplitude
- $FiO_2 < 70\%$, $PaO_2 > 60$, good inflation: wean MAP
- Extubation: term MAP 11–14, amp ≤ 20 ; LBW MAP 10–12, ≤ 15 ; VLBW MAP < 10 , < 12
- Air leak: wean MAP hard, smallest amplitude, accept CO_2 high 50s–low 60s, PIE side down

$$OI = MAP \times FiO_2 \times 100 \div PaO_2 \quad \text{Dir.}$$

OI > 40 for > 2 h, or in 3 of 5 gases: 80% predicted mortality (director's sheet).



Sources

Course: Dr. Kira Short's N5537 Intro to Neonatal FEN and Neonatal TPN lectures; FEN formulas handout; chorio and EOS handout. Your study guides: Cardiovascular Function & Treatments; Choosing a Pressor; Neonatal Hemodynamics; Module 2 respiratory; Blood Gas Interpretation (Pat Thomas).

Card: your instructor's badge card. **Dir.:** program director's NNP Brain.

Bilirubin: AAP 2022 Clinical Practice Guideline (Kemper et al.); Maisels et al., J Perinatol 2012; UNC NCCC hyperbilirubinemia guideline (2025).

BP: Dionne, Abitbol, Flynn 2012 (erratum) via Batton, Clin Perinatol 2020; Starship and Medscape neonatal hypertension.

Vent: Loyola neonatal ventilation supplement; SPA oscillator review; UNC NCCC high-frequency ventilation; UW Madison NICU jet guideline; NHS GGC HFOV.

Bicarb: Canterbury DHB and NSW ANMF neonatal guidelines.

The full versions, calculators, and links are in the NNP app.